

Energy-Efficiency-Based Joint Uplink Resources Allocation for LEO Satellite Beam-Hopping System

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Abstract—Multidimensional resources management enhances the efficiency of low-Earth orbit (LEO) satellite systems by dynamically allocating wireless resources. However, the majority of existing studies have overlooked the evolutionary trend of terrestrial terminals (TTs) from homogeneous to heterogeneous types. For instance, energy efficiency (EE) requirements for TTs inherently vary due to differences in their operating environments and deployment costs. Thus, this article investigates the design of beam hopping (BH) patterns, frequency allocation, and power control for LEO satellite uplinks, considering the diverse EE sensitivity (EES) among TTs. Specifically, a resource management problem is formulated to maximize the weighted sum EE, which captures the differences in EES among TTs. Next, an undirected graph is constructed based on EES and spatial isolation, and matching algorithms are proposed according to the many-to-one matching theory to obtain BH patterns. After that, to solve the joint frequency allocation and power control problem, a penalty function is applied to handle the binary variable utilized for expressing frequency allocation. Finally, quadratic transform and minorize–maximization are employed to concave the original problem, which guarantees to obtain a suboptimal solution. Simulation results demonstrate that proposed BH design methods provide a higher optimization ceiling for joint resources management than relevant benchmarks. Besides, the proposed uplink resources management significantly improves the EE by 34% in LEO satellite uplink compared with ignoring EES scheme. This work establishes a novel paradigm for energy-efficient uplink resource management in heterogeneous LEO systems.

Index Terms—Low-Earth orbit (LEO), many-to-one matching, minorize–maximization (MM), uplink resources management, weighted sum energy efficiency (WSEE).

I. INTRODUCTION

AS LOW-EARTH orbit (LEO) satellite systems improve by leaps and bounds, the next generation LEO satellite requires greater communication capacity to meet the

rapidly increasing demands for communication [1]. Thus, the multibeam technique is utilized by the LEO system. The freedom in time and space of beams is significantly increased, benefiting from the use of this technology. However, this advantage gives rise to growing complexity in the management of multidimensional resources (i.e., frequency, power, time), and the traditional methods (e.g., preassigned) already cannot satisfy the needs of LEO in terms of communication capacity, efficiency, etc. [2]. In this situation, flexible multidimensional resource allocation attracts more and more focus and investigation.

Existing resources management technologies include beam hopping (BH) [3], power control, frequency allocation [4], precoding [5], etc. Applying these methods, the satellite has significantly enhanced overall energy efficiency, communication capacity, and anti-jamming capabilities. Furthermore, the joint optimization of the technologies mentioned above is carried out to achieve better performance [4], [6], [7], [8], [9]. It is worth noting that most of the current research focuses on enhancing downlink performance, while relatively little attention is paid to the uplink. Although some downlink methods are transferable to uplink, the exponential growth of terrestrial terminals (TTs) (e.g., the number of Starlink users has surpassed 4 million) in LEO constellations (e.g., Starlink [10], Kuiper, Telesat [11]) and the diversification of TT types (fixed terminals, Internet of Things (IoT) terminals [12], and Internet of Vehicles (IoV) terminals [13]) introduce new challenges. Due to variations in operational environments and deployment costs, different types of terminals exhibit heterogeneity. For instance, the energy efficiency sensitivity (EES) among different TT types exhibit significant discrepancies. Compared with fixed terminals, IoT and IoV exhibit higher EES [13], [14]. This is because fixed terminal is typically powered by the grid, while IoT and IoV generally rely on battery power. This implies that these TTs, which are typically situated at a considerable distance from a continuous power source, desire more channel capacity per unit of energy consumption [12]. Consequently, in contrast to the homogeneity of satellite beams during the downlink [15], the TTs in the uplink exhibit heterogeneity. Existing downlink-centric methods have not considered such differences among the research objects and cannot adequately address uplink heterogeneity [7], [8], [16]. Furthermore, as the wireless communication field increasingly focuses on energy efficiency (EE) [17], [18], improving the EE of the uplink has become a crucial aspect in the development of high-efficiency satellite

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communication systems. Implementing uplink resource management for different types of TTs is essential to enhance EE. Hence, with the growing diversity of TT types and the increasing focus on EE, the uplink resource management must account for the differences in EE sensitivity (EES) among TTs to improve the EE of the uplink LEO satellite systems.

A. Related Works

To enhance the performance of resources management, conventional BH approaches have mitigated the mutual interference by controlling the distance among active beams. Chen et al. [5] designed the BH patterns to reduce the number of precoding operations while satisfying the requirements. In [3], a matching algorithm was proposed to minimize the gap between traffic demand and offered capacity for downlink. Moreover, Hu et al. [19] investigated BH patterns from maximizing downlink throughput and proposed BH schemes assisted by deep reinforcement learning (DRL). Similarly, Lin et al. [20] proposed a multiagent DRL method to select the BH patterns. To mitigate interference among beams, they imposed a requirement of adjacent beam avoidance. Furthermore, frequency and power are crucial wireless resources in satellite communication systems. The joint optimization of frequency allocation and power control has already been conducted [6], [7], [8], [9]. Specifically, a DRL-based resource allocation algorithm was proposed in [6] to maximize the number of users and system throughput joint optimization goal in downlink. Wang et al. [7] allocated the frequency segments and power to mitigate interference by employing Majorization–Minimization algorithm. Di et al. [8] formulated the optimization problem as a many-to-many two-sided matching game to maximize the sum rate and solve power control and frequency allocation iteratively. Lv et al. [9] proposed a distributed iterative algorithm based on the mechanism of simulated annealing (SA) to solve multiobjective problems. Maximizing throughput may lead to rapid battery drainage for TTs. Evidently, most current resource management schemes focus on achieving the traffic demands but ignore the differences in EES among TTs.

To maximize the EE, Zappone et al. [21] showed the differences among various forms of EE, such as global EE (GEE) and weighted sum EE (WSEE). The objective of [15], [22], [23], and [24] is to maximize GEE (MaxGEE). Yang et al. [15] proposed a method to iteratively optimize the BH and precoding by two-step quadratic transform in satellite downlink. In [22], a method based on successive convex approximation was adopted to solve energy-efficient beamforming in the user-centric network. Khalili et al. [23] proposed a method based on Majorization–Minimization to maximize the EE by joint design of subchannel assignment, power control, and antenna selection. Hashemi et al. [24] reformulated the EE by applying the monotonic optimization and solved it by polyblock outer approximation method. Although GEE can represent the efficiency of the entire network with the strongest physical interpretation, it limits the tuning resources of individual links based on their EES. Venturino et al. [25] employed WSEE and let the weights

account for the priority of the scheduled users in downlink. This demonstrates that WSEE facilitates the association of weights with link characteristics.

B. Motivations and Contributions

As previously discussed, with the rapid development of LEO satellite communication, the types of TTs have shown a trend toward diversification. This diversification may lead to divergent priorities in uplink and downlink resource management for LEO satellite systems. Previous studies that focus on traffic demands, load balancing, or interbeam interference have failed to address the unique requirements of the uplink, where different TT types impose varying demands on EE. Moreover, the EE of the uplink will emerge as a pivotal metric for evaluating the satellite system performance. Therefore, inspired by [25], to better account for the EES of TTs compared with GEE, we apply WSEE to ensure that terminals with higher EES can achieve greater EE during resource allocation. Consequently, this article investigates uplink resource management for maximizing the WSEE in LEO satellite systems.

The main contributions of this article are summarized as follows.

- 1) To investigate the resource management of the LEO satellite uplink from the perspective of the transformation of TT types from homogeneous to heterogeneous, we build the models of BH pattern design, frequency allocation, and power control for the uplink of the LEO satellite system and formulate a resource management problem to maximize the WSEE of system, which makes it possible to tune the resources of TTs based on their EES.
- 2) Considering the EES variations of terminals and the interference among beams, the BH pattern design is formulated as a many-to-one matching problem. We construct an undirected graph based on EES and spatial isolation to describe peer effects among beams. Then, two beams-BH patterns matching algorithms (BBHMAs) are proposed, which provide a larger optimization space for power control and frequency allocation to maximize EE.
- 3) We reduce the problem's complexity and propose a method called joint power and frequency algorithm (JPFA) to jointly optimize power and frequency to address the mixed-integer nonlinear program (MINLP) problem. This method maximizes the system EE and, meanwhile, reduces the inter-TT interference and meets the traffic demands according to the differences in EES of different TT types. This method also guarantees to obtain an efficient solution in polynomial time with acceptable performance.
- 4) Simulation results demonstrate the advantages of considering EES in BBHMA and JPFA methods compared with the conventional benchmarks. These results also prove the effectiveness of enhancing EE to manage resources for the LEO satellite uplink according to the

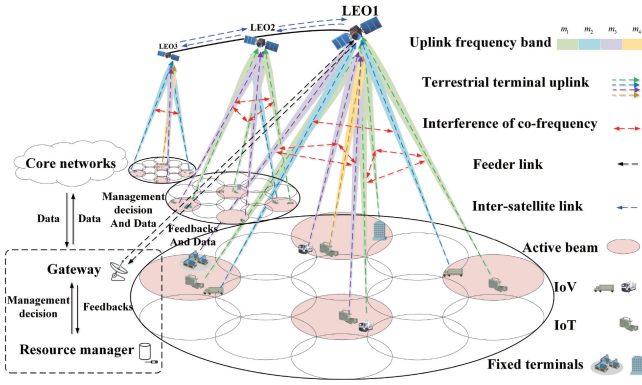


Fig. 1. Illustrative scenario of the uplink LEO satellite BH system, the TTs which are located at illuminated beams will receive the co-frequency interference of interbeams.

differences in EES of TTs. Proposed methods provide a novel paradigm for uplink resource management.

The remainder of this article is organized as follows. In Section II, we describe the uplink system model and formulate the WSEE maximize problem. In Section III, we decompose the primal problem into BH patterns design and joint resources allocation. We decouple them as a many-to-one matching game with peer effect and fractional programming based on quadratic transform and minorize-maximization (MM), respectively. In Section IV, we evaluate and discuss the performance of the proposed algorithms compared with benchmarks. In Section V, we conclude this article.

II. SYSTEM MODEL AND PROBLEM FORMULATION

In this article, we take the uplink of the LEO satellites BH system into account as shown in Fig. 1. The LEO satellites travel along their predefined orbital paths. A LEO satellite can cover an area S that can be divided into J cells, and utilizes BH technology to serve TTs due to limited wireless resources. Similar to the downlink communication process [3], the uplink process includes the following procedures: First, users' statuses (such as user ID, traffic demand, channel conditions, users' positions and so on) are sent to the gateway by feeder links. Then, the resource manager deployed at the gateway executes the resource management algorithm and makes decisions of beam pattern design, power control and frequency allocation based on users' statuses. Subsequently, the gateway transmits decisions to the satellites via feeder links. After decoding these decisions, the satellites transmit the uplink control information through downlink. Next, users send traffic data to the satellites, and finally the satellites deliver the data to the core network.

A LEO can generate K beams at one time slot t_{slot} , which represents the minimum service time. Satellite takes time T to traverse all the cells at least once. We refer to the active beams at one slot t_{slot} as a "BH pattern" and T is the BH period. Let us represent $\mathbf{A}_t = [a_1^t, a_2^t, \dots, a_j^t, \dots, a_J^t]$ as a BH pattern at slot t ; the element $a_j^t \in \{0, 1\}$ means the status of beam j , which is a binary assignment variable, and where $a_j^t = 1$ (or $a_j^t = 0$) means beam j is active (or inactive) at slot t . There are n_{user}^t TTs in K beams at time slot t which are represented

by set L_{user}^t , L_j^t represents the set of TTs served by j_{th} beam at time slot t and the number of elements in this set is n_j^t , which satisfies $\bigcap_{a_j \in A_t} L_j^t = \emptyset$ and $\bigcup_{a_j \in A_t} L_j^t = L_{\text{user}}^t$. Furthermore, each beam shall be active at least once within a BH period.

A. Channel Gain Model

Let $|\mathbf{H}|^2 \in \mathbb{C}^{K \times n_{\text{user}}^t}$ denote the uplink channel gain matrix which indicates the n_{user}^t TTs to the multibeam satellite at time slot t , the element can be expressed as $|h_{j,l}|^2$ which is channel gain from the l_{th} TT to the j_{th} beam. The element $|h_{j,l}|^2$ can be written as [3], [5]

$$|h_{j,l}|^2 = \frac{G_{R_{j,l}} G_{T_l}}{L_{\text{loss}}} \left(\frac{\lambda}{4\pi d_{kl}} \right)^2. \quad (1)$$

In (1), $G_{R_{j,l}}$ is the receiving antenna gain of the j_{th} beam at the l_{th} TT location. The location information will be reported to the satellite by the TT. G_{T_l} is the transmit antenna gain of l_{th} terminal. d_{kl} is the distance between l_{th} TT and the satellite, L_{loss} represents sum of other losses (e.g., atmospheric path loss, shadowing margin, scintillation loss). Besides, λ represents the wavelength of uplink. In uplink communication, frequency offset precompensation and timing advance are performed among TTs [26]. Therefore, the Doppler shift is not included in our channel gain model, and we will discuss the impacts of residual frequency offsets in Section IV-F.

B. Frequency Allocation and Power Control Model

We assume full frequency reuse among beams. The TTs in one cell can be served by frequency-division multiple access (FDMA) [27]. The uplink bandwidth of a beam is expressed as F_{total} , which can be divided into at most M parts, the least allocated bandwidth is $F_{\text{least}} = F_{\text{total}}/M$. Then, let $\mathbf{F}^{j,l,t}$ denote the allocation bandwidth of l_{th} TT which is served by j_{th} beam at time slot t [16], it can be expressed as

$$\mathbf{F}^{j,l,t} = [f_1^{j,l,t}, f_2^{j,l,t}, \dots, f_m^{j,l,t}, \dots, f_M^{j,l,t}]. \quad (2)$$

And $f_m^{j,l,t} \in \{0, 1\}$, when $f_m^{j,l,t}$ is equal to 1, it means the m_{th} bandwidth being allocated to terminal l . To avoid interference within the beam, the spectrum of the beam can only be assigned to one TT at a time.

Next, let P_{max}^l denote the maximum transmission signal power of the l_{th} TT. This information will be known when terminals access the network. In order to eliminate mutual interference among beams, there are some methods, such as beamforming [28], precoding, and power control. Considering that LEO can serve a huge area (e.g., Starlink can cover more than 0.64 million square kilometers at 550km orbit), the number of TTs is a massive order of magnitude. So, to ease the computing pressure on ground and reduce the consumption of signaling information (e.g., downlink control information), we utilize a power control method to reduce the interference effect rather than beamforming or precoding. The power control can be designed as $w_m^{j,l,t}$, it denotes the power control coefficient of l_{th} TT at m_{th} band which can be sent by the satellite, if it is equal to 0, it means this band is not allocated to l_{th}

terminal, $0 \leq \sum_{m=1}^M f_m^{j,l,t} w_m^{j,l,t} \leq 1$. The transmission power of l_{th} terminal at slot t can be expressed as

$$P_l^t = P_{\max}^l \sum_{m=1}^M f_m^{j,l,t} w_m^{j,l,t} \quad (3)$$

when $\sum_{m=1}^M f_m^{j,l,t} w_m^{j,l,t} = 1$, it represents the l_{th} TT working at maximum transmission power.

It is noteworthy that our main focus lies in the design of BH patterns and the resource management among users rather than on the association between cells and satellites. Meanwhile, in the uplink scenario, due to the strong directionality of the terminal antenna and the considerable distance among the LEO satellites, the satellite experiences minimal uplink interference from intersatellite users. Additionally, TTs within the same beam employ the FDMA, so there is no inter-TT interference among TTs within the same beam. Therefore, we only consider inter-TT interference within the satellite in subsequent research. Then, the receiving signal to interference plus noise ratio (SINR) of l_{th} TT which is served by beam j in band m at slot t can be expressed as

$$\text{SINR}_m^{j,l,t} = \frac{a_j^t f_m^{j,l,t} w_m^{j,l,t} P_{\max}^l |h_{j,l}|^2}{\sigma^2 + \sum_{a_l \in A_t, u \neq j} \sum_{v \in L_u^t} a_l^t f_m^{u,v,t} w_m^{u,v,t} P_{\max}^v |h_{j,v}|^2} \quad (4)$$

where u is the beam that is illuminated simultaneously with j and v is the TT served by u . $|h_{j,v}|^2$ is the channel gain from v_{th} TT to j_{th} beam. $\sigma^2 = F_{\text{least}} \kappa T_{nt}$, which is the variance of the additive white Gaussian noise (AWGN). κ , T_{nt} represent the Boltzmann's constant and the noise temperature of receive beams, respectively. The AWGN of each beam in F_{least} band is independent and identically distributed, i.e., $n \sim N(0, \sigma^2)$. Then, the actual uplink communication capacity of l_{th} TT can be written as

$$C^{j,l,t} = a_j^t \sum_{m=1}^M f_m^{j,l,t} F_{\text{least}} \log_2(1 + \text{SINR}_m^{j,l,t}). \quad (5)$$

C. Problem Formulation

In this section, we formulate an optimization problem to maximize the WSEE of uplink by optimizing beam pattern design, power control, and frequency allocation. The uplink EE of l_{th} terminal can be expressed as

$$E^{j,l,t} = \frac{C^{j,l,t}}{P_l^t + P_{\text{sys}}} \quad (6)$$

where P_l^t denotes the transmission power and P_{sys} denotes the power to maintain normal operation of the ground user, such as power supply and circuit blocks [29].

Next, we consider that different terminals have varying sensitivities to EE, i.e., IoT terminals are often battery-powered and deployed in energy-constrained environments (e.g., smart sensors), necessitating a high EES to prioritize long-term operation. Given the stable power sources, such as vehicle batteries, EE is of secondary importance for IoV terminals, and thus a lower EES is justified. Fixed terminals (e.g., Starlink fixed sites) are often grid-powered, allowing even lower EES values as energy conservation is less critical.

To avoid increasing the network burden due to additional signaling overhead, we assume that the core network stores the EES of various terminals. When the terminals access the network [30], the EES information can be identified by the Network Access ID of the terminal in the resource manager. Thus, we introduce a weight factor λ_l for l_{th} terminal to represent its EES when allocating resources. We assume the value of factor is in $(0, \xi]$, three different types of TTs are classified into three levels. The range of fixed terminals is from 0 to $(1/3)\xi$, IoV is in $((1/3)\xi, (2/3)\xi]$ and IoT is from $(2/3)\xi$ to ξ . A larger value of λ_l implies that, during the optimization process, there will be a greater inclination to enable TT to achieve higher EE.

Then, we utilize the WSEE to represent the EE of the LEO satellite in uplink transmission at slot t . it can emphasize the different sensitivity between terminals compared to GEE, and it can be written as

$$E_{\text{total}}^t = \sum_{a_j^t \in A_t} \sum_{l \in L_j^t} \lambda_l E^{j,l,t}. \quad (7)$$

To maximize the EE in a BH period T , we construct an objective optimization problem for time–frequency–power domain allocation, the problem is formulated as $\mathcal{P}_0$ in (8), let A , F , and w denote the sets of A_t , $F^{j,l,t}$ and $w_m^{j,l,t}$. They, respectively, represent the BH patterns, the uplink frequency domain resources allocation, and the coefficient of transmission power control of terminal l . In constraint C1, after the resources management of satellites, the capacity of terminal l must satisfy its minimum-rate requirement. Constraints C2 and C3 indicate the range of TT's power control coefficient. Constraint C4 implies that one bandwidth can only be allocated to one TT in an arbitrary beam, and the sum of bandwidths allocated to the TTs in beam j is no more than the total bandwidth. In constraint C5, the number of active beams is limited per slot t , and we assume that a cell can be active at least once per period T . Constraints C6 and C7 indicate that the frequency allocation and BH patterns are restricted to binary values. In constraint C8, a TT can be served by only one beam

$$\begin{aligned} (\mathcal{P}_0) : & \max_{A, F, w} \sum_{t=1}^T E_{\text{total}}^t & (8) \\ \text{s.t.} & C1 : C^{j,l,t} \geq C_{l_min_need,t} & \forall l \forall t \forall j \\ & C2 : \sum_{m=1}^M f_m^{j,l,t} w_m^{j,l,t} \leq 1 & \forall l \forall t \forall j \\ & C3 : 1 \geq w_m^{j,l,t} \geq 0 & \forall l \forall t \forall m \\ & C4 : \sum_{l \in L_j^t} \|F^{j,l,t}\|_1 \leq M, \sum_{l \in L_j^t} f_m^{j,l,t} = 1, & \forall j \forall t \\ & C5 : \|A^t\|_1 = K, \sum_{t=1}^T a_j^t \geq 1 & \forall j \forall t \\ & C6 : f_m^{j,l,t} \in \{0, 1\} & \forall j \forall l \forall t \\ & C7 : a_j^t \in \{0, 1\} & \forall j \forall t \\ & C8 : \bigcap_{a_j \in A_t} L_j^t = \emptyset & \forall t. \end{aligned}$$

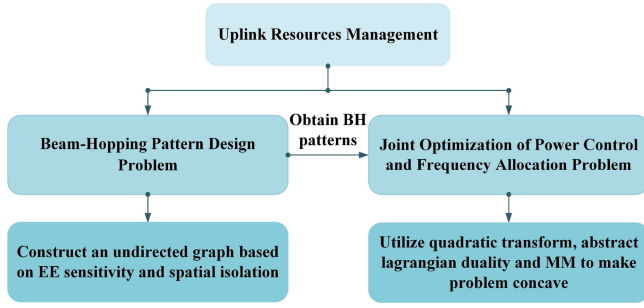


Fig. 2. LEO satellite system uplink resources management problem decomposition architecture.

$\mathcal{P}_0$ consists of a fractional form of the channel capacity divided by the total power consumption. It is a sum-of-ratio problem. Due to the nonconvexity of SINR [31], this is also a Concave-convex fractional programming (CCFP) problem. Meanwhile, the other variables $\mathbf{A}_t$ and $\mathbf{F}^{j,l,t}$ are multidimensional variables composed of binary variables, thus, this is a MINLP problem, which is an NP-hard problem in general. So, we cannot obtain the global optimal solution in polynomial time.

III. PROBLEM REFORMULATION AND PROPOSED SCHEMES

In this section, in order to solve the uplink resources management $\mathcal{P}_0$, we naturally divide $\mathcal{P}_0$ into two subproblems: the BH pattern design problem and the joint optimization of power control and frequency allocation problem. The structure is illustrated in Fig. 2. Then we, respectively, propose two algorithms to deal with these two subproblems.

A. Many-to-One Matching Game for Beam Hopping Pattern Design

To develop an algorithm for solving the problem of uplink BH pattern design based on maximizing EE, we can consider beams and BH patterns as two finite and disjoint sets, $\mathbf{B} = \{b_1, b_2, \dots, b_J\}$ and $\mathbf{A} = \{\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_T\}$. The TTs within beams submit the request to satellites for resources and report their location information, thus, beams and BH patterns have complete information about each one. The integer problem program can be viewed as a many-to-one matching problem [32]. Let matching ρ describe the assignment of beams to BH patterns such that a beam can be matched to only one BH pattern, while a BH pattern can be matched to multiple beams. More formally

Definition 1: ρ is mapping the set $\mathbf{B} \cup \mathbf{A} \cup \{0\}$ to the set $\mathbf{B} \cup \mathbf{A} \cup \{0\}$, where $\rho(b_j) = \{\mathbf{A}_t \in \alpha : (b_j, \mathbf{A}_t) \in \rho\}$ and $\rho(\mathbf{A}_t) = \{b_j \in \mathbf{B} : (b_j, \mathbf{A}_t) \in \rho\}$, it is satisfied the following conditions. 1) $\rho(b_j) \subseteq \mathbf{A}$; 2) $\rho(\mathbf{A}_t) \subseteq \mathbf{B}$; 3) $|\rho(b_j)| = 1$; 4) $|\rho(\mathbf{A}_t)| = K$; and 5) $b_j \in \rho(\mathbf{A}_t) \Leftrightarrow \rho(b_j) = \mathbf{A}_t$.

The conditions 1) and 2) imply the matching relationship between beams and BH patterns. Condition 3) and 4) mean that the beam can only be active once with BH period T and a BH pattern can serve K beams, it indicates that no beams are inactive. K is the number of maximized active beams. It is worth mentioning that we can define multiple beams with

the same location in set $\mathbf{B}$, for some beams that cannot meet the requirements with a single illumination.

Observing $\mathcal{P}_0$ from the perspective of maximizing EE, it is obvious that the less interference the terminals experience, the higher the upper bound of EE is and large weighted terminals need to be emphasized. The terminals are bound to beams, and the distance among them determines the interference among beams. Beams within a BH pattern always tend to be far away from each other, this is the externalities (i.e., peer effects) of the matching problem [32], when externalities are considered, designing matching mechanisms is significantly more challenging [33], the existence of stable matchings has been difficult in the presence of externalities.

Then, to quantify the peer effect among beams, we define the “Friendship network.” This network among beams is modeled by a weighted graph $G(\mathbf{B})$, which includes two important indicators: EES weights and distance. The weights of the connections between nodes denote the relationships among beams. The strength of relationship between beams b_{j_1} and b_{j_2} can be represented by $I_{b_{j_1}, b_{j_2}}$

$$\lambda_{b_j} = \sum_{l \in L_j^i} \lambda_l \quad (9)$$

$$I_{b_{j_1}, b_{j_2}} = \begin{cases} 3\lambda_{\max} D & d_{b_{j_1}, b_{j_2}} \geq 3D \\ \lambda_{\max} d_{b_{j_1}, b_{j_2}} & D < d_{b_{j_1}, b_{j_2}} < 3D \\ -\nu D \lambda_{\max} & D \geq d_{b_{j_1}, b_{j_2}} \end{cases} \quad (10)$$

$$\lambda_{\max} = \max(\lambda_{b_{j_1}}, \lambda_{b_{j_2}}).$$

In (10), let the sum of weights of terminals EES represent the beams EES. As the beam distance increases, there is a greater preference between beams, and $\lambda_{\max}$ ensures that a beam with higher EES is farther away from the other beams and also ensures that the graph is undirected (i.e., $I_{b_{j_1}, b_{j_2}} = I_{b_{j_2}, b_{j_1}}$), ν is used as a penalty factor to indicate that adjacent beams repel each other, because the mutual interference between cells activated in the same time slot will be extremely severe when the distance between these cells is no greater than D . Thus, the value of penalty factor ν can be as large as possible to completely prevent adjacent cells from being active simultaneously. 3-D is three times the diameter of beams, and we will show the influence of different distances in Section IV-C. Taking the 3-D case as an example, we define that $I_{b_{j_1}, b_{j_2}}$ remains fixed when the distance between beams is bigger than 3-D. The purpose of this action is to prevent the preference of high EES beams from becoming too large, thereby ignoring the preferences of low EES beams

$$P_{b_j}(\rho) = \sum_{b_{j_1} \in \rho(\rho(b_j)), b_{j_1} \neq b_j} I_{b_j, b_{j_1}} \quad (11a)$$

$$P_{\mathbf{A}_t}(\rho) = \sum_{b_{j_1} \in \mathbf{A}_t} \sum_{b_{j_2} \in \mathbf{A}_t, b_{j_2} \neq b_{j_1}} I_{b_{j_1}, b_{j_2}} \quad (11b)$$

Preference lists of the beams and BH patterns: Despite the players of this matching being beams and BH patterns, the decision-making relies on satellites, and the players' information is known to each other by satellites. First, we introduce the utility functions of players, due to BH patterns

and beams without complementarities, beams only consider being far away from others within a BH pattern in (11a), and BH patterns want beams to “like” each other as much as possible in (11b). Then, we define the preference relation $\succ$ for beams and BH patterns. Specifically, for any beams $b_j \in B$, any two matchings $\rho, \rho', \rho(b_j) = A_{t_1}, \rho'(b_j) = A_{t_2}$, the preference of b_j is $\succ_{b_j}$, it can be described as

$$(A_{t_1}, \rho) \succ_{b_j} (A_{t_2}, \rho') \Leftrightarrow P_{b_j}(\rho) > P_{b_j}(\rho') \quad (12)$$

it indicates that b_j prefers A_{t_1} than A_{t_2} . Similarly, for any BH pattern $A_t \in \alpha$, any two matching ρ, ρ' and two beams $b_{j_1}, b_{j_2}, b_{j_1} \in \rho(A_t), b_{j_2} \in \rho'(A_t)$, other elements of $\rho(A_t)$ and $\rho'(A_t)$ are identical. The preference of A_t is $\succ_{A_t}$, it can be explained by

$$(b_{j_1}, \rho) \succ_{A_t} (b_{j_2}, \rho') \Leftrightarrow P_{A_t}(\rho) > P_{A_t}(\rho') \quad (13)$$

it implies that A_t prefers b_{j_1} than b_{j_2} .

Then, to obtain stable matchings, we introduce the swap matchings. Due to the preferences of BH patterns are determined by the composition of beams, thus, beams are free to exchange with other beams for moving to preferred BH patterns, we define a swap matching $\rho_{b_{j_1} b_{j_2}}^{b_{j_1}}$ which indicates that b_{j_1} and b_{j_2} exchange their BH patterns A_{t_1} and A_{t_2} [8] while keeping all other beams' matching the same, i.e., ρ is the previous matching, swap matching

$$\rho_{b_{j_1} b_{j_2}}^{b_{j_1}} = \{\rho \setminus (b_{j_1}, A_{t_1}), (b_{j_2}, A_{t_2})\} \cup \{(b_{j_1}, A_{t_2}), (b_{j_2}, A_{t_1})\}. \quad (14)$$

Next, we define the two-sided exchange-stable (2ES) if and only if no matching satisfies the following conditions

$$\begin{aligned} \forall e \in \{b_{j_1}, \rho(b_{j_1}), b_{j_2}, \rho(b_{j_2})\}, P_e(\rho_{b_{j_1} b_{j_2}}^{b_{j_1}}) &\geq P_e(\rho) \\ \exists e \in \{b_{j_1}, \rho(b_{j_1}), b_{j_2}, \rho(b_{j_2})\}, P_e(\rho_{b_{j_1} b_{j_2}}^{b_{j_1}}) &> P_e(\rho). \end{aligned} \quad (15)$$

It indicates that the values of BH patterns and beams utility functions are not decreasing, and there is at least one that increases, this is the necessary condition for swap matching $\rho_{b_{j_1} b_{j_2}}^{b_{j_1}}$ to be allowed.

Corollary 1: For a swap matching $\rho_{b_{j_1} b_{j_2}}^{b_{j_1}}$, if $P_{b_{j_1}}(\rho_{b_{j_1} b_{j_2}}^{b_{j_1}}) \geq P_{b_{j_1}}(\rho)$ and $P_{b_{j_2}}(\rho_{b_{j_1} b_{j_2}}^{b_{j_1}}) \geq P_{b_{j_2}}(\rho)$, as long as b_{j_1} and b_{j_2} propose the swap matching, it will be approved. this benefits from the undirected graph of the “friendship network.”

With the definition of 2ES, we develop two BBHMA for finding 2ES matching, the BBHMA-1 consists of the initial matching phase and swap matching phase. In step 1, we utilize the improved Gale–Shapley algorithm for the many-to-one matching problem to obtain an initial matching, this matching may be stable, but not optimal. In step 2, the satellite searches for a swap matching within the current matching. When a swap matching satisfies Corollary 1, the swap matching will be approved and replace the current matching, and the loop will terminate when the matching is in the 2ES.

It is worth noting that even if the BBHMA-1 can obtain a 2ES matching, the result may not be optimal solution, because it lacks the ability to escape from local optimum due to its greedy strategy. Thus, we utilize the SA method to improve

Algorithm 1 Beam-BH Pattern Matching Algorithms-1

Step 1:Initial Matching Phase.

- 1: When the BH patterns are empty, randomly selects one beam and matches it with the BH pattern.
- 2: **while** BH patterns are not full **do**
- 3: Idle beams simultaneously submit applications for BH patterns based on their preferences.
- 4: BH pattern A_t receives requests from various beams (e.g., b_j, b_i), A_t prefers b_j than b_i based on its preferences, i.e., $(b_j, \rho) \succ_{A_t} (b_i, \rho')$.
- 5: Beam b_j and the beams within BH pattern A_t form a new pattern together.
- 6: **end while**
- 7: Obtain the current matching ρ

Step 2:Swap Matching Phase.

- 8: **while** The satellite searches the current matching ρ and can find any beams b_j and b_i that form a swap matching which satisfies Corollary 1. **do**
- 9: Approve the swap matching $\rho_{b_j b_i}^{b_j}$, in which b_j and b_i exchange their BH patterns with each other.
- 10: Set $\rho = \rho_{b_j b_i}^{b_j}$.
- 11: **end while**

Output: The current matching ρ which is 2ES.

End of algorithm

the matching algorithm. In BBHMA-2, we still adopt step 1 in BBHMA-1 to initial the matching phase. For modeling the process of SA, we define the approving probability of swap matching in (17), T is the “current temperature.” When a swap matching satisfies Corollary 1, it will always be approved. Conversely, whether to accept the match is determined by probability P_T

$$\Delta P = (P_{b_{j_1}}(\rho_{b_{j_1} b_{j_2}}^{b_{j_1}}) + P_{b_{j_2}}(\rho_{b_{j_1} b_{j_2}}^{b_{j_1}})) - (P_{b_{j_1}}(\rho) + P_{b_{j_2}}(\rho)) \quad (16)$$

$$P_T = \begin{cases} 1, & \text{if } \rho_{b_{j_1} b_{j_2}}^{b_{j_1}} \text{ is approved} \\ \exp(\frac{\Delta P}{T}), & \text{if } \rho_{b_{j_1} b_{j_2}}^{b_{j_1}} \text{ isn't approved.} \end{cases} \quad (17)$$

T_{ini} is the initial temperature, τ is the attenuation coefficient of temperature, i_{max} is the Markov chain, those are key parameters to BBHMA-2, The closer τ is to 1, the larger i_{max} and T_{ini} are, the greater the search range of the algorithm, and the more likely it is to find the optimal solution. However, the time consumption is also greater. Thus, it is necessary to find a balance between time consumption and performance.

1) Stability and Convergence Analysis: We now prove the stability and convergence of the proposed algorithms BBHMA. First, In the Step 2 of Algorithm 1, any beams b_{j_1} and b_{j_2} can apply a swap matching $\rho_{b_{j_1} b_{j_2}}^{b_{j_1}}$ as long as they meet Corollary 1, i.e., the preference of each beam after the exchange is greater than or equal to that before the exchange. Meanwhile, because of the undirected nature of the “friendship network,” the preference of each BH pattern can also be maintained without decreasing. It should be noted that, given the limited number of beams, the number of feasible swap matchings is also restricted. Consequently, there always exists a scenario in which, after a number of swap matchings are accepted, no further swap matchings can be found to ensure

Algorithm 2 Beam-BH Pattern Matching Algorithms-2**Step 1: Initial Matching Phase.**

- 1: Adopt step 1 in BBHMA-1 to initial the matching phase.
- 2: Obtain the current matching ρ

Step 2: Swap Matching Phase.

- 3: $T = T_{ini}$
- 4: **while** $T \geq T_{end}$ **do**
- 5: **while** $i \leq i_{max}$ **do**
- 6: **if** The satellite searches the current matching ρ and can find any beams b_j and b_i that form a swap matching which satisfies Corollary 1. **then**
- 7: Approve the swap matching $\rho_{b_j}^{b_i}$, in which b_j and b_i exchange their BH patterns with each other.
- 8: Set $\rho = \rho_{b_j}^{b_i}$.
- 9: **else**
- 10: Use the equation (35) to calculate the P_T , and set $\rho = \rho_{b_j}^{b_i}$ with probability P_T .
- 11: **end if**
- 12: Set $i = i + 1$
- 13: **end while**
- 14: Set $T = \tau \times T$ and $i = 0$
- 15: **end while**

Output: The current matching ρ .

End of algorithm

that the preferences of each beam and pattern do not decrease. At this point, the BBHMA algorithm converges to the final matching ρ^* , which is in the 2ES state and represents a stable solution. Second, Häggström [34] and Henderson et al. [35] have proved SA approach can find the global optimum with an adequately large number of iterations. Thus, the convergence of Algorithm 2, which is based on SA, is not always considered [8], because it is a two-loop structure that constrained by the initial temperature T_{ini} and the length of Markov chain i_{max} .

2) *Computational Complexity Analysis*: Next, we can discuss the complexity of BBHMA. For initialing matching phase, it primarily consists of two parts: preference calculation and sorting, as well as delayed acceptance, the former is the major consumption part that its complexity is $O(J^2)$. In step 2 of BBHMA-1, the satellite first needs to search for feasible swap matching, the complexity of this procedure is $O(J(J - K))$ which includes the searching process and calculating utility functions according to Corollary 1. It is difficult to evaluate when the matching become 2ES, thus, we assume that the matching is 2ES after I iterations, the complexity of BBHMA-1 is $O(IJ(J - K))$. In BBHMA-2, the complexity is associated with i_{max} , τ and T , which is usually extremely high, so we limit the size of parameters to control complexity in experiments.

B. Power Allocation With Fixed Integer Solution

After resolving the BH problem, there still exists an integer programming problem regarding frequency allocation. While BH patterns are determined, the EE is independent between

different BH patterns, $F^{j,l,t}$ and $w^{j,l,t}$ are updated at each slot. Thus, we focus on the t th slot in the following work, and the subscript/superscript t is omitted for simplicity. To avoid integer programming, we suppose the frequency allocation is fixed, $F^{j,l}$ is known. Thus, the remaining problem is about power allocation, which can be written as $\mathcal{P}_1$

$$\begin{aligned}
 (\mathcal{P}_1) : & \max_{\mathbf{w}} E_{total} \\
 \text{s.t.} \quad & C1' : C^{j,l} \geq C_{l_min_need} \quad \forall l, \forall j \\
 & C2' : \sum_{m=1}^M f_m^{j,l} w_m^{j,l} \leq 1 \quad \forall l \forall m \forall j \\
 & C3' : w_m^{j,l} \geq 0 \quad \forall l \forall m \\
 & C8 : \bigcap_{a_j \in A_t} L_j^t = \emptyset \quad \forall t.
 \end{aligned} \tag{18}$$

When $F^{j,l}$ is fixed, the interference among terminals can be expressed directly. However, $\mathcal{P}_1$ is still nonconcave because of the nonconvexity of SINR. Observing this problem, it is a sum-of-ratio and fractional problem. One of the approaches to address this problem are decoupling the numerator and denominator, and reformulate it become a concave problem, e.g., dinkelbach's transform, and quadratic transform [31]. Quadratic transform has shown advantage over dinkelbach's transform in multiratio fractional programming by keeping equivalent objective after reformulating the original FP. SINR can be written as $\eta_m^{j,l}$ through quadratic transform

$$\eta_m^{j,l} = 2\theta_m^{j,l} \sqrt{a_{jl}^{j,l} w_m^{j,l} P_{max}^l |h_{j,l}|} - \left(\theta_m^{j,l} \right)^2 * \left[\sigma^2 + \sum_{a_u \in A_t, u \neq j} \sum_{v \in L_u^t} a_{ul}^{u,v} w_m^{u,v} P_{max}^v |h_{j,v}|^2 \right]. \tag{19}$$

Then, by utilizing a quadratic transform, the objective function EE can be expressed as follows:

$$\begin{aligned}
 Y_{total} = & \sum_{a_j \in A_t} \sum_{l \in L_j} \left(2\mu_l \sqrt{\lambda_l \sum_{m=1}^M F_{least} \log_2 (1 + \eta_m^{j,l})} \right. \\
 & \left. - \mu_l^2 (P_l + P_{sys}) \right).
 \end{aligned} \tag{20}$$

Thus, we convert the E_{total} from fractional format to Y_{total} , and the constraints C1 can be written as

$$\sum_{m=1}^M f_m^{j,l} F_{least} \log_2 (1 + \eta_m^{j,l}) \geq C_{l_min_need} \quad \forall l. \tag{21}$$

When we fix the $\theta_m^{j,l}$ and μ_l , Y_{total} is a concave function. The problem can be written as

$$\begin{aligned}
 (\mathcal{P}_2) : & \max_{\mathbf{w}, \theta_m^{j,l}, \mu_l} Y_{total} \\
 \text{s.t.} \quad & (21), C2', C3', C8.
 \end{aligned} \tag{22}$$

When $\theta_m^{j,l}$ and μ_l is fixed, $\mathcal{P}_2$ is a concave problem. Then, we can utilize the iterative approach to optimize $w_m^{j,l}$ with fixed $\theta_m^{j,l}$ and μ_l which can be updated under fixed $w_m^{j,l}$. The solution of $\mathcal{P}_2$ is summarized in Algorithm 3. In iteration, $\theta_m^{j,l}$ and μ_l

Algorithm 3 Approach for Power Controlling**Input:** $F_{\text{least}}, \mathbf{F}^{j,l,t}, |h_{j,l}|^2, \mathbf{A}_t$.

- 1: Initialize $\mathbf{w}$ randomly via $\mathcal{P}_2$
- 2: **repeat**
- 3: Update $\theta_m^{j,l}$ and $\mu_l^{j,l}$ via (23) and (24).
- 4: Update $w_m^{j,l}$ by solving $\mathcal{P}_2$.
- 5: **until** convergence

Output: $\mathbf{w}$

are updated with fixed power control and $w_m^{j,l}$ is optimized by $\theta_m^{j,l}$ and $\mu_l^{j,l}$ where come from

$$\theta_m^{j,l} = \frac{\sqrt{a_{j,l} f_m^{j,l} w_m^{j,l} P_{\max}^l |h_{j,l}|}}{\left[\sigma^2 + \sum_{a_u \in \mathbf{A}_t, u \neq j} \sum_{v \in L_u^l} a_u f_m^{u,v} w_m^{u,v} P_{\max}^v |h_{j,v}|^2 \right]} \quad (23)$$

$$\mu_l^{j,l} = \frac{\sqrt{\lambda_l \sum_{m=1}^M f_m^{j,l} F_{\text{least}} \log_2(1 + \text{SINR}_m^{j,l})}}{\left(\sum_{m=1}^M f_m^{j,l} w_m^{j,l} P_{\max}^l + P_{\text{sys}} \right)}. \quad (24)$$

$\mathcal{P}_2$ can be solved by CVX which utilizes interior-point method. When the difference between the current EE and the previous EE is less than a certain value τ , e.g., $\tau = 10^{-2}$, the current w_l is a stationary point of the problem. Then we analyze the complexity of Algorithm 3, it primarily stems from line 4 which utilizes the interior-point method with the complexity of $O((Mn)^{3.5} \log(1/\varepsilon))$, where Mn is the size of optimal variable, $n = n_{\text{user}}^l$ and M is the number of bands in Algorithm 3, and $\varepsilon > 0$ is the duality gap. Thus, the complexity of Algorithm 3 is $O(N(Mn)^{3.5} \log(1/\varepsilon))$, N is the maximum number of iterations because of “repeat.”

C. Joint Optimization of Frequency Allocation With Power Control

In the previous section, we have proposed the method to transform the nonconcave to concave with the fixed frequency allocation. Thus, in this section, we deal with the non concave resulting from the multiplication of $f_m^{j,l}$ and $w_m^{j,l}$. First, we rewrite constraint C2

$$C2 : \left\{ \sum_{m=1}^M f_m^{j,l} w_m^{j,l} \leq 1, \Rightarrow \begin{cases} f_m^{j,l} = 0 \Rightarrow w_m^{j,l} = 0 \\ f_m^{j,l} = 1 \Rightarrow w_m^{j,l} \leq f_m^{j,l} \\ \sum_{m=1}^M w_m^{j,l} \leq 1. \end{cases} \right. \quad (25)$$

Specifically, while $f_m^{j,l}$ is equal to 0, the l_{th} terminal does not require allocating power at m_{th} band. So, the new C2 is represented as

$$\begin{aligned} C2_a : 0 \leq w_m^{j,l} \leq f_m^{j,l} \\ C2_b : \sum_{m=1}^M w_m^{j,l} \leq 1. \end{aligned} \quad (26)$$

Moreover, frequency allocation variables $f_m^{j,l}$ are binary, which leads to the objective function being MINLP. To handle the

binary variable $f_m^{j,l}$, we relax it to be a real value between 0 and 1, and rewrite the constraint C6 as follows:

$$C6_a : f_m^{j,l} \in [0, 1] \quad \forall j \quad \forall l. \quad (27)$$

For forcing $b_m^{j,l}$ the relaxed variables to take close to 0 or 1, we define a new constraint $C6_b$

$$C6_b : \sum_{\forall j} \sum_{l \in L_j^l} \sum_{m=1}^M \left(f_m^{j,l} - \left(f_m^{j,l} \right)^2 \right) \leq 0. \quad (28)$$

$C6_b$ limits the value of $f_m^{j,l}$ equal to 0 or 1. Then, we organize the above constraints, the channel capacity $C^{j,l}$ can be denoted as

$$C^{j,l} = \sum_{m=1}^M F_{\text{least}} \log_2 \left(1 + \frac{w_m^{j,l} P_{\max}^l |h_{j,l}|^2}{\sigma^2 + \sum_{a_u \in \mathbf{A}_t, u \neq j} \sum_{v \in L_u^l} a_u w_m^{u,v} P_{\max}^v |h_{j,v}|^2} \right). \quad (29)$$

Next, the new optimization problem can be written as

$$\begin{aligned} (\mathcal{P}_3): \max_{\mathbf{w}, \mathbf{F}} E_{\text{total}}(\mathbf{w}) &= \sum_{a_j \in \mathbf{A}_t} \sum_{l \in L_j^l} a_j \lambda_l \frac{C^{j,l}}{\sum_{m=1}^M w_m^{j,l} P_{\max}^l + P_{\text{sys}}} \\ \text{s.t. } C4': \sum_{l \in L_j^l} \sum_{m=1}^M f_m^{j,l} &\leq M, \sum_{l \in L_j^l} f_m^{j,l} = 1 \quad \forall j \\ C1', C2_a, C2_b, C6_a, C6_b, C8. \end{aligned} \quad (30)$$

In $\mathcal{P}_3$, the objective function $E_{\text{total}}(\mathbf{w})$ can be transformed to concave $\bar{E}_{\text{total}}(\mathbf{w})$ by the quadratic transform in Algorithm 3, constraint C1' can also transformed to C1'' by utilizing this method like (21). In constraint $C6_b$, it is not concave. We use the abstract Lagrangian duality, add $C6_b$ into the objective function as a penalty term. So, the abstract Lagrangian function of $\mathcal{P}_3$ can be represented as

$$\begin{aligned} L(\mathbf{w}, \mathbf{F}) &= \bar{E}_{\text{total}}(\mathbf{w}) \\ &\quad - \zeta \sum_{\forall j} \sum_{l \in L_j^l} \sum_{m=1}^M \left(f_m^{j,l} - \left(f_m^{j,l} \right)^2 \right). \end{aligned} \quad (31)$$

ζ is the penalty factor to penalize the variable $b_m^{j,l}$ when the value is not close to binary numbers. The problem of $\mathcal{P}_4$ can written as

$$\begin{aligned} (\mathcal{P}_4): \max_{\mathbf{w}, \mathbf{F}} L(\mathbf{w}, \mathbf{F}) \\ \text{s.t. } C4', C1'', C2_a, C2_b, C6_a, C8. \end{aligned} \quad (32)$$

Proposition 1: When penalty factor ζ is large enough, The solution of problem $\mathcal{P}_4$ is feasible only when $\mathbf{F}$ belongs to the set $\{0, 1\}$, thus, the problem $\mathcal{P}_4$ is equivalent to the problem $\mathcal{P}_3$.

Proof: The proof is given in the Appendix. ■

Algorithm 4 Joint Power and Frequency-Based MM

Input: $F_{\text{least}}, h_{j,l}, i = 0, \mathbf{w}(0), \mathbf{F}(0), |h_{j,l}|^2, \zeta, \mu_l, \theta_m^{j,l}$.

- 1: $\mathbf{w}(0), \mathbf{F}(0)$: the feasible value of $w_m^{j,l}$ and $b_m^{j,l}$ which satisfy all of constraints at the initial iteration
- 2: i, I_{max} : the current iteration index and the maximum number of iterations
- 3: ζ : the penalty factor which is larger than 1
- 4: $\mu_l, \theta_m^{j,l}$: the parameters of Quadratic transform
- 5: **repeat**
- 6: Update the μ_l and $\theta_m^{j,l}$ via (23) and (24) by employing $\mathbf{w}(i)$ and $\mathbf{F}(i)$. Then, update $C1''$ via (21) and obtain $\bar{E}_{\text{total}}(\mathbf{w})$.
- 7: Calculate $\nabla_{\mathbf{F}} L_2(\mathbf{F}(i))$ and obtain $\tilde{L}(\mathbf{w}, \mathbf{F})$.
- 8: Set $i = i + 1$.
- 9: Solve $\mathcal{P}_5$ by utilizing CVX and obtain $\mathbf{w}(i)$ and $\mathbf{F}(i)$.
- 10: **until** $i = I_{\text{max}}$ or convergence.

Output: $\mathbf{w}(i)$ and $\mathbf{F}(i)$

After abstract Lagrangian, the problem of $\mathcal{P}_4$ is a nonlinear programming (NLP), and we redefine it as follows:

$$L(\mathbf{w}, \mathbf{F}) = L_1(\mathbf{w}, \mathbf{F}) - L_2(\mathbf{F}) \quad (33)$$

$$L_1(\mathbf{w}, \mathbf{F}) = \bar{E}_{\text{total}}(\mathbf{w}) - \zeta \sum_{\forall j} \sum_{l \in L_j^t} \sum_{m=1}^M f_m^{j,l} \quad (34)$$

$$L_2(\mathbf{F}) = -\zeta \sum_{\forall j} \sum_{l \in L_j^t} \sum_{m=1}^M \left(f_m^{j,l} \right)^2. \quad (35)$$

Although both $L_1(\mathbf{w}, \mathbf{F})$ and $L_2(\mathbf{F})$ are concave, the concavity of the objective function $L(\mathbf{w}, \mathbf{F})$ is hard to verify, because it is the difference between two concave functions, this belongs to DC programming. In order to obtain the concave approximation of $L(\mathbf{w}, \mathbf{F})$, which is contributed to design an efficient iterative algorithm for obtaining a local optimum, we utilize the MM to get a tight lower band for the optimization $\mathcal{P}_4$ [36]. As $L_2(\mathbf{F})$ is concave, it satisfies the first order condition for concavity [37], therefore we use first-order Taylor expansion to obtain $\tilde{L}_2(\mathbf{F})$

$$\begin{aligned} \tilde{L}_2(\mathbf{F}) &= L_2(\mathbf{F}(i-1)) + \nabla_{\mathbf{F}} L_2(\mathbf{F}(i-1)) \\ &\quad \times (\mathbf{F}(i) - \mathbf{F}(i-1)) \end{aligned} \quad (36)$$

where $\mathbf{F}(i-1)$ is the result of $\mathcal{P}_4$ at $(i-1)$ th iteration, and $\nabla_{\mathbf{F}} L_2$ is the first-order derivative of $L_2(\mathbf{F})$ with respect to $\mathbf{F}$. $L_2(\mathbf{F}) \leq \tilde{L}_2(\mathbf{F})$, and $\tilde{L}_2(\mathbf{F})$ can make a tight lower bound of $\mathcal{P}_4$. After applying MM Algorithm, the objective function is represented as

$$\tilde{L}(\mathbf{w}, \mathbf{F}) = L_1(\mathbf{w}, \mathbf{F}) - \tilde{L}_2(\mathbf{F}) \quad (37)$$

which is always concave. We rewrite $\mathcal{P}_4$ as follows:

$$\begin{aligned} (\mathcal{P}_5): \quad & \max_{\mathbf{w}, \mathbf{F}} \tilde{L}(\mathbf{w}, \mathbf{F}) \\ \text{s.t.} \quad & C4', C1'', C2_a, C2_b, C6_a, C8. \end{aligned} \quad (38)$$

The objective function and constraints of $\mathcal{P}_5$ are a concave optimization problem that can be solved by the optimization tools (e.g., CVX), the detailed process of the JPFA is shown in

Algorithm 4. Where line 6 is the process of quadratic transform to make the problem and some constraints concave. In line 7, we utilize the abstract Lagrangian duality to treat the binary variable $f_m^{j,l}$, and make lagrangian function concave by the MM algorithm. Then, $\mathcal{P}_5$ can be solved by CVX in line 9.

1) *Stability and Convergence Analysis:* The objective function of $\mathcal{P}_5$ and $\mathcal{P}_4$ have the following relations:

$$\begin{aligned} & L_1(\mathbf{w}(i+1), \mathbf{F}(i+1)) - L_2(\mathbf{F}(i+1)) \\ & \geq L_1(\mathbf{w}(i+1), \mathbf{F}(i+1)) - L_2(\mathbf{F}(i)) \\ & \quad - \nabla_{\mathbf{F}} L_2(\mathbf{F}(i)) \times (\mathbf{F}(i+1) - \mathbf{F}(i)) \\ & = \max_{\mathbf{w}, \mathbf{F}} L_1(\mathbf{w}, \mathbf{F}) - L_2(\mathbf{F}(i)) \\ & \quad - \nabla_{\mathbf{F}} L_2(\mathbf{F}(i)) \times (\mathbf{F} - \mathbf{F}(i)) \\ & \geq L_1(\mathbf{w}(i), \mathbf{F}(i)) - L_2(\mathbf{F}(i)) \\ & \quad - \nabla_{\mathbf{F}} L_2(\mathbf{F}(i)) \times (\mathbf{F} - \mathbf{F}(i)) \\ & = L_1(\mathbf{w}(i), \mathbf{F}(i)) - L_2(\mathbf{F}(i)). \end{aligned} \quad (39)$$

Therefore, when we solve $\mathcal{P}_5$ by the proposed algorithm JPFA, each iteration guarantees that the objective function is monotonically nondecreasing. It ensures that a larger value can always be achieved in each iteration as long as the iteration does not stop. Thus, a locally optimal solution can be obtained. We define convergence threshold κ . When the change amount of the objective function between adjacent iterations is less than κ , we consider that the algorithm has completed convergence at $(i+1)$ th iteration

$$\kappa \geq |\tilde{L}(\mathbf{w}(i+1), \mathbf{F}(i+1)) - \tilde{L}(\mathbf{w}(i), \mathbf{F}(i))|. \quad (40)$$

2) *Initial Solution and Computational Complexity Analysis:* Note that a random initial solution would suffer from a serious premature turning-off issue [31], thus appropriate initial values of w_m^l and $f_m^{j,l}$ are critical for the objective function. In order to obtain a feasible solution, we assume that the interferences between beams are set as a fixed parameter which can be seen as maximum interference. Then, the complexity of this problem is decreased from the beam pattern at any slot to the beam. Thanks to the simplification of the problem, the nonconvexity of SINR no longer needs to be considered. Thus, a quadratic transform can be applied once, and the integer and noninteger variables can be directly solved jointly using a solver (e.g., MOSEK). After the above operations, a suitable initial solution can be obtained. Next, we analyze the complexity of Algorithm 4, the main complexity comes from line 9. The optimization problem $\mathcal{P}_5$ consists of $2nM$ variables. The numbers of both the spectrum allocation variables and the power control variables are nM . Here, n represents the total number of TTs, and M denotes the number of available spectra for allocation. $\varepsilon > 0$ is also the quality gap, so, the complexity of Algorithm 4 is $O(I_{\text{max}}(2nM)^{3.5} \log(1/\varepsilon))$ which is polynomial time complexity.

IV. SIMULATION RESULTS AND PERFORMANCE ANALYSIS

A. Simulation Settings and Comparison Methods

In this section, we evaluate the performance of the proposed uplink BH algorithms and the algorithm joint power control and frequency allocation in a LEO satellite BH system. The parameter settings of the simulation are summarized in Table I.

TABLE I
MAJOR PARAMETERS

Parameters	Values
The height of the satellite	1200km
3dB angle θ_b	1.7647°
Diameter of beams D	40km
Frequency band for uplink f_c	30GHz
Total bandwidth B_{total}	400MHz
The minimum bandwidth B_{least}	50MHz
Satellite maximum receive gain G_{max}	38.5dBi
Number of beams J	144
Number of maximum active beams K	9
Number of terminals per beam	4
Satellite receive antenna pattern	Refer to 3GPP TR 38.811
Timeslot duration t	2ms (Refer to DVB-S2X)
The BH period T	$16 \times t$

We consider a LEO satellite to be deployed at an altitude of 1200 km, it can serve 144 cells, and the maximum number of active beams is equal to 9 at each slot t which is set to 2 ms according to DVB-S2X [38]. Thus, the period of BH T is 32 ms. The uplink frequency is reused among beams, and the total bandwidth is 400MHz which can be divided into eight equal parts. The equivalent satellite receiving antenna aperture d is equal to 0.33 m, and the radiation pattern which refers to 3GPP TR 38.811 [39] is considered as follows:

$$G_R(\theta) = \begin{cases} G_{max}, & \theta = 0^\circ \\ G_{max} + 10\log_{10}4 \left| \frac{J_1(\pi d \sin(\theta)/\lambda)}{\pi d \sin(\theta)/\lambda} \right|^2, & 0^\circ < |\theta| \leq 90^\circ \end{cases} \quad (41)$$

where λ is the wavelength of uplink frequency, θ represents the off-axis angle in degrees, and J_1 is the Bessel function of the first kind and first order. G_{max} is the maximum gain of the satellite receiving antenna.

We consider three types of TTs: 1) IoT terminals; 2) IoV terminals; and 3) fixed terminals. In terms of EES, IoT terminals are the largest, followed by IoV, with fixed terminals being the smallest. It should be noted that a degree of EES variation is to be expected within any of the broad categories of terminals, so their values belong to [4, 6), [2, 4), (0, 2), respectively. We deploy TTs to simulate scenarios following 2-D normal distributions [40]. The quantities of various types of TTs are allocated in the order of more fixed terminals than IoV and more IoV than IoT, which is in line with the gradual evolution trend of the satellite market. Additionally, we deploy four TTs in each beam which referred to [23] and [41], and a single satellite covers a total of 576 TTs. Furthermore, their uplink traffic demands D_i always are less than downlink demands [42]. Therefore, we assume that the uplink traffic demands are not strongly correlated with terminal types and are generated uniformly at random, i.e., $D_i \in [10r\text{Mbps}, 50r\text{Mbps}]$, where r is the scale factor of the traffic demand $r \in \{0.5, 0.6, \dots, 1.5\}$, and $r = 0.5$ means the low demand and $r = 1.5$ implies high demand [5], specifically. It is important to note that while this article investigates various types of terminals, the primary concern is the EES, and thus, we consider the parameters of each type to be the same in most simulations (e.g., maximum transmit antenna gain 43.2 dBi, maximum transmit power 3 dBW), we also simulate and analyze the JPFA performance at different

transmit powers in Section IV-E to ensure that the simulation is comprehensive. Transmit antennas of TTs are always aimed at the satellite, which relies on ephemeris based on satellite broadcasting.

In order to investigate the benefits of Algorithm BBHMA which we propose to solve the BH pattern design to maximize the WSEE, we compare it with the following conventional BH methods.

- 1) *The Random BH (R-BH)*: The satellite randomly chooses 9 beams to be active at each slot, and the beams will be active once after 16 slots. In other words, the random process precludes the reactivation of the same beam, every cell will be served.
- 2) *The Spatial Isolation BH (SP-BH)*: The satellite chooses the active beams that are far away from each other with a fixed distance, note that the excessive distance setting will increase the period for activating all beams, and thus, so we choose an appropriate distance to ensure the period T .
- 3) *The Periodic BH (P-BH)*: The satellite takes a fixed BH pattern in every period T , which means that each beam will be active periodically.
- 4) *The DRL-Based BH (DRL-BH)*: The BH scheduling, as presented in [20], is based on DRL and enforces the adjacent beam avoidance requirement to reduce the action space.

The selected benchmarks remain widely adopted in satellite resource management owing to their interpretability and low computational latency. Moreover, recent references [7], [20] also indicate that these methods can serve as foundational baselines. We consider the TTs within a beam as an ultraterminal and the EES of the beam is the sum of terminals without resource management when we conduct the simulation of BH patterns design. In addition, SP-BH-2-D and SP-BH-3-D indicate that the distance among beams at any slot is equal to twice and three times the beam diameter, respectively, while four times the beam diameter cannot generate BH patterns at the same period T .

To evaluate the performance of JPFA, we will utilize the BH patterns obtained by the proposed BBHMA-1 when not specifically mentioned, and we will compare the performance achieved by the proposed JPFA method with the following benchmark methods.

- 1) *The Fixed Allocation (FA)*: The satellite allocates the frequency bands fixedly, and the power in each assigned band is obtained by the proposed Algorithm 3 which is based on quadratic transform.
- 2) *MaxGEE*: The approach proposed in [23] aims to maximize the GEE, which has no concern with the EES, and the approach is based on dinkelbach method and MM.
- 3) *Maximize Channel Capacity (MaxCC)*: The objective function is to maximize the total capacity of BH patterns [7], and utilizes quadratic transform to process SINR, then Algorithm 4 is adopted to optimize power control and frequency allocation. Additionally, since the problem is a nonconvex mixed-integer programming problem, obtaining its optimal solution is

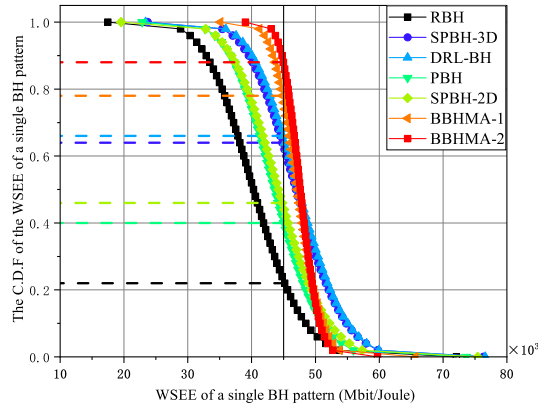


Fig. 3. Comparison of the proposed BBHMA methods with conventional BH methods in terms of the C.D.F of the WSEE of a single BH pattern.

computationally intractable. To this end, to quantify the solution quality, we assume that interference cancellation can be performed among the beams, that is, there is no interference among the beams. A known upper-bound (UP-bound) is then defined based on this assumption.

- 4) *UP-bound*: We assume that the satellite can rely on SIC to eliminate the interference between beams, this leads to the complexity of the optimization problem being further reduced. Then, it can be solved by the MOSEK after quadratic transform.

B. Performance Analysis for Uplink BH Patterns Design

For performance comparison of the BH methods, we conduct the simulations in various regions to evaluate the proposed methods. Thus, we randomly generated over 2000 LEO satellite communication snapshots. Then, with regard to the simulation parameters, the penalty factor ν in (10) is equal to 1000 and the “initial temperature” of BBHMA-2 T_{ini} is set as 2500, the attenuation coefficient of temperature τ is equal to 0.97, i_{max} is set as 10000. Furthermore, the performance metric is illustrated by the cumulative distribution function (C.D.F) of the WSEE of BH patterns at any slot t or period T . Fig. 3 shows the C.D.F of E_{total}^t the WSEE of a single BH pattern at any slot $\Pr(E_{\text{total}}^t \geq E)$. We assume that the value of $\tilde{E}$ 45 000¹ is set as a base value which is the minimum tolerance for WSEE and shown as a black solid line in Fig. 3. It is observed that the C.D.F of $\Pr(E_{\text{total}}^t \geq 45,000)$ arranges according to the following order: R-BH < P-BH < SP-BH-2-D < SP-BH-3-D < DRL-BH < BBHMA-1 < BBHMA-2. Our proposed methods outperform the conventional method by at least 15% in terms of the C.D.F, and they make EE of the beam pattern more concentrated. Notably, compared with the SPBH-3-D algorithm, the DRL-BH algorithm achieves only a 2% performance improvement at the minimum tolerance for WSEE. The reason lies in the fact that the DRL-BH mitigates interference in a manner similar to that of SPBH-3-D, still relying on distance isolation. The performance improvement

¹This value can be calculated by using the reference SINR and the average EES obtained from the distribution of the number of terminals and the EES numerical range in the current snapshot.

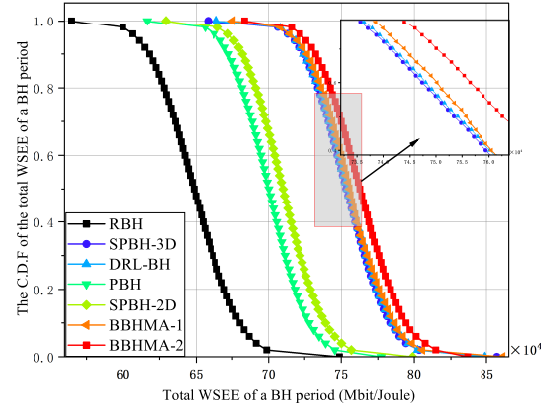


Fig. 4. Comparison of the proposed BBHMA methods with conventional BH methods in terms of the C.D.F of the total WSEE of a BH period.

is attributed to the search process in reinforcement learning. Meanwhile, the performance of the DRL-BH is inferior to that of the BBHMA. The reason lies in the fact that DRL overlooks the differences among cells in terms of EES during the exploration process.

Note that when the value of WSEE is larger than 50 000, the C.D.F of the proposed methods begins to intersect with that of conventional methods, and the reason behind this is that the conventional BH pattern design only focus on the distance between active beams, while ignoring their EES, thus, there will be high-EES clustered in a BH pattern; the proposed methods take into account both distance and EES, the sensitive beam will be given special care during the process of matching, which benefits from building stronger “friendship” (10) with other beams, and the sensitive beam will be scattered into different patterns to enhance the “friendship.” Therefore, there are intersection points. In addition, the $\Pr(E_{\text{total}}^t \geq 45\,000)$ of BBHMA-2 is larger than BBHMA-1, this benefits from the SA operation of the matching process, enabling to escape from the local optimum.

To evaluate the performance more comprehensively, Fig. 4 shows the C.D.F of E_{total}^T , which is the total WSEE of BH period, $\Pr(E_{\text{total}}^T \geq E)$. Our proposed methods demonstrate superior performance compared with conventional ones. Regarding BBHMA-1, SPBH-3-D, and DRL-BH, their results do not show significant differences. This is because the selected metric, which is the cumulative sum of WSEE across all beams during a BH period, inherently obscures optimization at the time-slot level. However, the proposed algorithm BBHMA-1 can offer a larger optimization space for the subsequent JPFA in comparison to SPBH-3-D and DRL-BH. This will be verified in Section IV-D.

In addition to generating 2000 snapshots to verify the impact of the algorithm’s performance on different spatial distributions, we conducted simulations on the performance of the proposed methods for different quantity distributions of terminals. In the simulation, we set up four cases to examine the impact of terminals with different quantity distributions on the algorithm, as follows: 1) $\text{IoT} > \text{IoV} > \text{FX}$; 2) $\text{IoV} > \text{FX} > \text{IoT}$; 3) $\text{FX} = \text{IoV} = \text{IoT}$; and 4) $\text{FX} > \text{IoV} > \text{IoT}$. As depicted in Fig. 5, it can be observed that different quantity

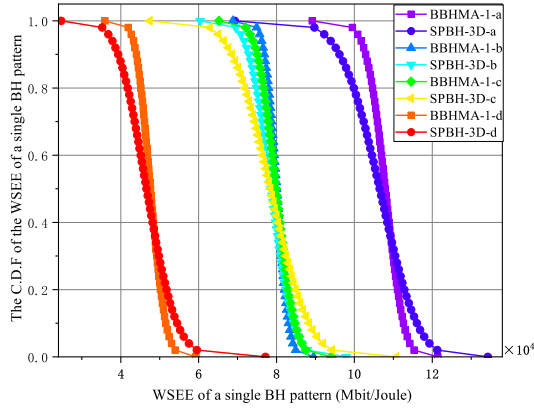


Fig. 5. Comparison of BBHMA-1 method and SP-3-D method under different quantity distributions of TTs in terms of the C.D.F of the WSEE of a single BH period.

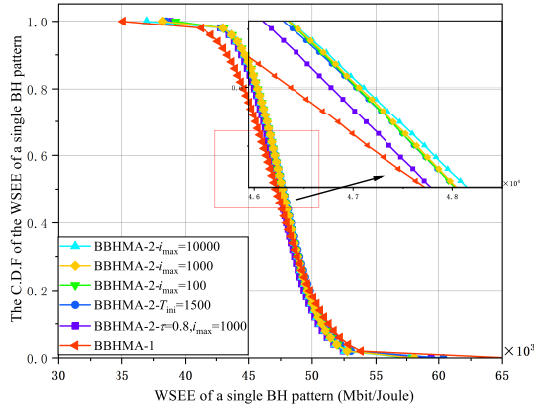


Fig. 6. Performance comparison of BBHMA-2 (based on SA strategy) under different parameter configurations with respect to the C.D.F of WSEE in a single BH pattern.

distributions will obtain different WSEE numerical ranges, yet the proposed method can still ensure superiority over existing algorithms.

To comprehensively evaluate the impact of various parameters on the proposed algorithm, we vary some parameter values to obtain different performances. As depicted in Fig. 6, it can be seen that when $\tau = 0.8$ and $i_{\max} = 1000$, the performance is the worst compared with that of the other four groups of parameters. This is because the excessively rapid temperature decrease prevents the algorithm from escaping the local solution. Meanwhile, it can also be observed that appropriately reducing the initial temperature T_{ini} and the value of $i_{\max}$, although causing some performance degradation, significantly reduces the computational complexity. For example, when $i_{\max}$ decreases from 10 000 to 1000, the computational complexity is reduced to one-tenth of the original. This is because the search space of our problem is not large, and a relatively small-scale search can find a local solution. Thus, it is acceptable to sacrifice part of the performance by adjusting parameters to reduce the algorithm complexity.

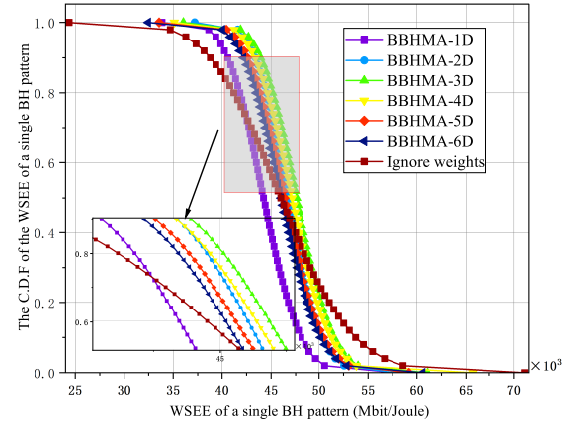


Fig. 7. Comparison of the different settings of proposed BBHMA methods regarding the C.D.F of the WSEE of a single BH pattern.

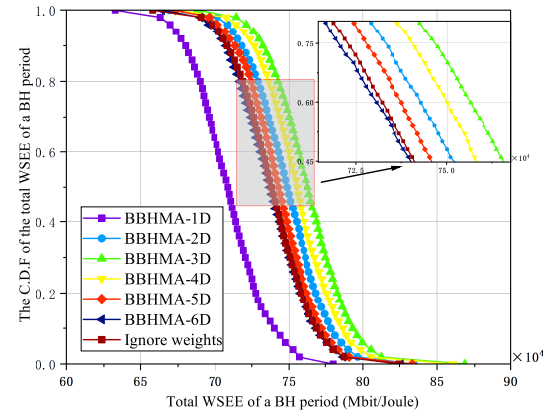


Fig. 8. Comparison of the different settings of proposed BBHMA methods regarding the C.D.F of the total WSEE of a BH period.

C. Influence of Distance and Energy Efficiency Sensitivity in Algorithm

Figs. 7 and 8 demonstrate the influence of the distance in (10) and EES on the proposed methods. Fig. 7 examines their impacts of different distance settings in (10) and ignoring EES on the WSEE of a BH pattern at any slot t . We set the distances as 1-D, 2-D, 3-D, 4-D, 5-D, and 6-D in the proposed method BBHMA-1. It is obvious to observe that the best performance is obtained when the distance is equal to 3-D from Fig. 7. Furthermore, with the distance increasing beyond 3-D, the performance will decline. This is because EE low-sensitivity beams will be ignored in the process of matching under the strong “friendship” of EE high-sensitivity beams. As expected, consistent with the occurrence of intersections in Fig. 3, the performance of BBHMA-1 degrades when we ignore the EES of beams in the matching process. This is because the BBHMA algorithm, which solely focuses on the interbeam distance and ignores EES, is essentially similar to the SPBH-3-D algorithm. When EES is taken into account, it prevents cells with large-EES from being allocated to the same BH time slot, enabling a reasonable distribution of EES values of different magnitudes in each BH time slot. Next, Fig. 8 shows the total WSEE of BH period T in the instances of different distances and ignoring EES, it is observed that

TABLE II
AVERAGE NUMBER OF SWAP OPERATIONS VERSUS DISTANCES

Distance in (10)	1D	2D	3D	4D	5D	6D
Number of swap operations	10	10	11	13	14	18

3-D can get larger WSEE compared with other distances, and ignoring EES will make WSEE degrade.

Meanwhile, the average number of swap operations versus different distance settings is shown in Table II, the number increases with the distance. According to the experimental results, the algorithmic complexity of the 6-D case is expected to be twice that of the 3-D case, owing to the expansion of the swap space. Thus, we conclude that the proposed methods in 3-D can get better performance than conventional methods, and for all subsequent simulations, we will utilize 3-D in (10).

D. Performance and Convergence Analysis for Joint Uplink Resources Allocation

First, to further evaluate the performance of the proposed BBHMA methods, we combine different BH methods with the proposed JPFA scheme. For choosing an appropriate value of penalty factor ζ in JPFA

$$\Delta f = \frac{\sum_{\forall j} \sum_{l \in L_j^i} \sum_{m=1}^M |f_m^{j,l} - \text{round}(f_m^{j,l})|}{Jn_j M} \quad (42)$$

the binary error is represented by Δf , which is calculated by (42). Herein, $\text{round}()$ represents rounding the data to either 0 or 1, and the denominator $Jn_j M$ is the total number of variables. As shown in Fig. 9 when the penalty factor ζ increase more than 800, the binary error, it can be reduced to the level of 1×10^{-8} . Therefore, we consider that when the penalty factor value is exceeds 800, $\mathcal{P}_4$ is equivalent to the $\mathcal{P}_3$.

Fig. 10 depicts the total WSEE of a BH period versus the number of iterations. It can be observed that the BH patterns obtained by BBHMA methods achieve the best performance compared with conventional BH methods since EES is introduced into the “friendship” network in BBHMA, allowing highly EES terminals to selectively combine with other terminals. This prevents the clustering of highly sensitive terminals and thereby expands the optimization space. In addition, benefiting from the ability to leapfrog to locally optimal solutions, the BBHMA-2 is better than BBHMA-1. We can also see, the effectiveness of JPFA for maximizing WSEE from Fig. 10. These results verify the conclusion that BBHMA is superior to the benchmark schemes in Section IV-B again.

Next, we will utilize BBHMA-1 in the following simulations because the complexity of BBHMA-1 is less than BBHMA-2. Fig. 11 shows the WSEE of a single BH pattern versus the number of iterations with the scale factor of the traffic demand being set to 1. As can be observed from Fig. 11, our proposed method converges to a stationary point after a finite number of iterations. The concavity properties derived in Section III-B guarantee the convergence of the

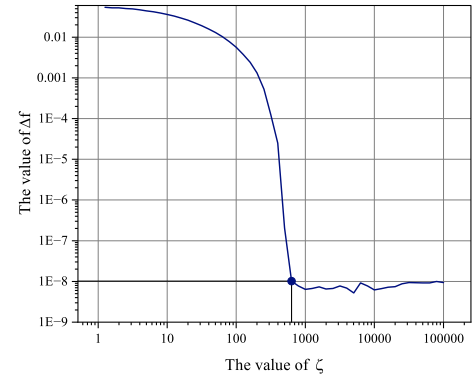


Fig. 9. Comparison of integer errors of frequency allocation parameters obtained by the proposed JPFA algorithm under different values of the penalty factor ζ .

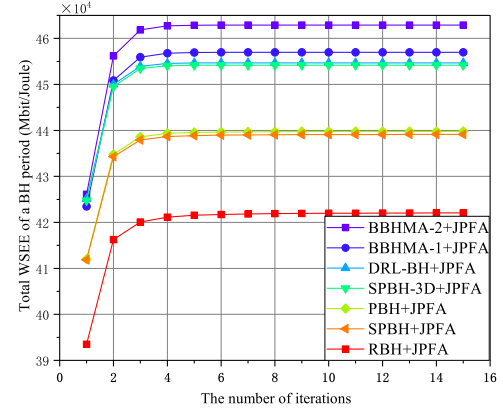


Fig. 10. WSEE of a BH period versus the number of iterations for the different BH methods with JPFA.

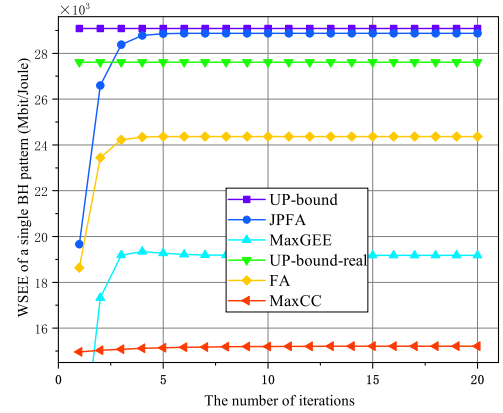


Fig. 11. WSEE of a single BH pattern slot versus the number of iterations for the proposed JPFA method and benchmarks, where scale factors of traffic demand are 1.

proposed JPFA based on interior-point method. Specifically, the relations between the objective function of $\mathcal{P}_5$ and $\mathcal{P}_4$ in (39) validates the convergence observed in Fig. 11. Thus, in the following simulations, we set the number of iterations as 10. The UP-bound-real values are calculated without interference cancellation based on the results obtained from the UP-bound. Its performance is lower than our proposed method because it ignores the interference among beams

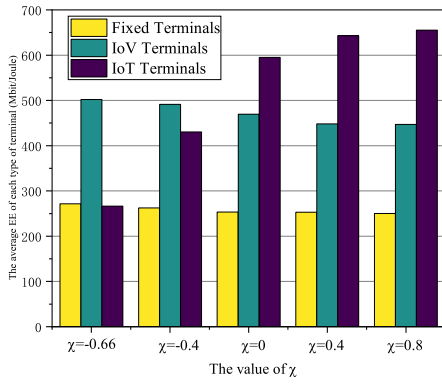


Fig. 12. Average EE of different types of terminals under different biases for proposed JPFA algorithm, where scale factors of traffic demand are 1.

when allocating frequency bands. In MaxGEE, it neglects the differences among terminals in the optimization process, resulting in WSEE not being optimized. For the worst line MaxCC, the resources are allocated to the terminals as many as possible and ignore the EE. Therefore, our proposed JPFA method can significantly improve the WSEE by 34% for the uplink LEO satellite system when compared with the case of ignoring the EES scheme. Meanwhile, it can guarantee convergence and achieve a result that approaches this UP-bound more closely in comparison with other algorithms.

Additionally, the numerical ranges of EES are $(0, \frac{1}{3}\xi]$, $(\frac{1}{3}\xi, \frac{2}{3}\xi]$ and $(\frac{2}{3}\xi, \xi]$, respectively, corresponding to fixed terminals, the IoV and the IoT ($\xi = 6$). Note that in the system, the resource allocation of various types of terminals is not determined by the absolute values of the EES. Instead, it is determined by the relative values of the EES of various types of terminals, that is, the ratio of the values. Thus, we add a bias $\chi\xi$ to adjust the ratio, and we define the ratio between Fixed terminals and IoT as

$$\zeta_{FX}^{\text{IoT}} = \frac{\chi\xi + (\frac{2}{3}\xi + \xi)}{(0 + \frac{1}{3}\xi)} = 3(\chi + 1), \left(\chi \geq -\frac{2}{3}\right) \quad (43)$$

we conducted sensitivity analyzes to evaluate how variations in EES values impact system performance. Fig. 12 shows the average EE of each type of terminal versus different bias χ . It should be noted that, to reflect the influence of the EES on the proposed algorithm, the ordinate EE considered herein is unweighted. By increasing the bias of the EES parameter for IoT terminals, the EE of this type of terminal has been significantly improved, and the EE of other types of terminals starts to decline. This is attributable to the fact that the proposed optimization algorithm rationally designs the hopping beam pattern, allocates frequency bands, and controls power according to the EES of the terminals. It is noted that the increase in the EE of IoT terminals in the Fig. 12 is not consistent with the decrease in the EE of other terminals. This is because the number distributions of various terminals in our simulation are not uniform ($FX > \text{IoV} > \text{IoT}$).

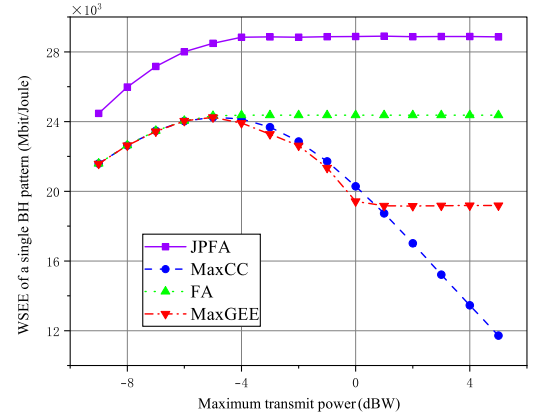


Fig. 13. WSEE of a single BH pattern versus different maximum transmit power for the proposed JPFA method and benchmarks, where scale factors of traffic demand are 1.

E. Influence of Different Maximum Transmit Power and Scale Factors of the Traffic Demand

The simulations presented above assume equal maximum transmit power for TTs (3dBW). However, this assumption is unrealistic in practice, since some types of TTs do not require such high power to meet their traffic demands. Thus, in this section, we compare the performance of different maximum transmit powers. Fig. 13 illustrates the WSEE of a single BH pattern versus the maximum transmit power for the proposed method and benchmarks with fixed scale factors of the traffic demand $r = 1$. As illustrated in this figure, we can observe that the performance of all methods exhibits an improvement until the transmit power reaches the value of -4dBW , this is because the numerator of the fractional function EE shows a dominant tendency to grow much faster than the denominator for intervals less than -4dBW . Furthermore, the gaps between different methods are mainly determined by the frequency allocation result. Then, with the increase of maximum transmit power, the WSEE of JPFA converges to a stationary point, and the performance of other methods deteriorates due to the mismatch of their objective functions. The WSEE of the FA and MaxGEE can also converge to a stationary point. Based on this simulation, we can conclude that under the identical beam pattern settings as this simulation, JPFA consistently obtains optimal results when the transmit power of each terminal is within the interval greater than -4dBW . Furthermore, the optimal results must be within the range of $[WSEE(P_{\min}), WSEE(P_{\max})]$,² because the larger transmit power terminals can adjust power to match the lowest TT.

Next, we also simulate the WSEE of a single BH pattern versus different scale factors of the traffic demand under different maximum transmit power. As illustrated in Fig. 14, it can be observed that the WSEE will increase with the maximum transmit power and subsequently converge to a constant value when the demand density factor is fixed. This further reaffirms the conclusions previously obtained from Fig. 13. Furthermore, as the scale factor of the traffic demand

² $WSEE(P)$ represents the curve of JPFA in Fig. 13, $P_{\max}$ and $P_{\min}$ represent the maximum and minimum maximum transmit power of the BH patterns, respectively.

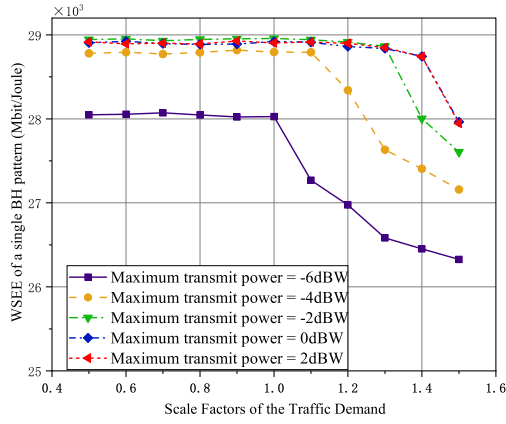


Fig. 14. WSEE of a single pattern slot versus different scale factors of the traffic demand for different maximum transmit power based on JPFA.

increases, the WSEE initially reaches a point of equilibrium and subsequently begins to decline. This is due to the fact that the demand is less than that provided by the stationary point of WSEE in the initial phase, as the scale factors increase, the point is no longer able to meet the demand, i.e., the point is outside the feasible domain. To address this, the capacity is boosted by changing the spectrum allocation as well as the power control during the joint optimization process. This leads to a reduction in the WSEE, nevertheless, when we observe the value of WSEE, we can find that it is still superior to the performance of the benchmarks presented in Fig. 13. All in all, our proposed method JPFA can obtain better performance than benchmarks with different maximum transmit powers and scale factors of the traffic demand.

F. Influence of Doppler Shift Compensation Errors and Terminal Mobility

Finally, to assess JPFA's robustness in real-world scenarios, we address terminal mobility by evaluating two critical aspects: residual Doppler frequency offset and terminal positioning errors. Specifically, Doppler effect represents one of the primary characteristics in satellite communication, and mobility management stands as one of the crucial issues to be addressed in satellite communication [43]. The residual frequency offset after precompensation, along with timing and other errors, can affect the decoding performance at the physical layer. At the network layer, this is manifested as a loss in channel capacity, that is, there will be a loss in the SINR. Therefore, to evaluate the impact of such errors, we conduct simulation analysis on the JPFA algorithm under different levels of SINR loss. We define the gap of WSEE which can be denoted as

$$\Delta WSEE_{sl} = WSEE_1(F_l, w_l) - WSEE_1(F_{nl}, w_{nl}) \quad (44)$$

F and w denote the frequency allocation and power control, respectively. The subscripts nl and l of these two variables indicate that the results are calculated by proposed JPFA algorithm under the condition of no SINR loss or with SINR loss. $WSEE_1()$ denotes the scenario with SINR loss. As depicted in

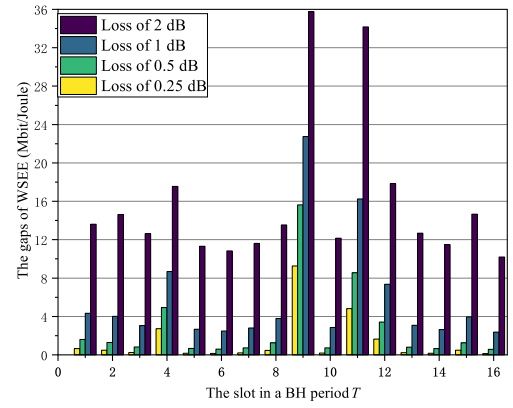


Fig. 15. Gaps in WSEE between the scenarios without SINR loss and those with SINR loss in a BH period T , where scale factors of traffic demand are 1 and maximum transmit power is 3 dBW.

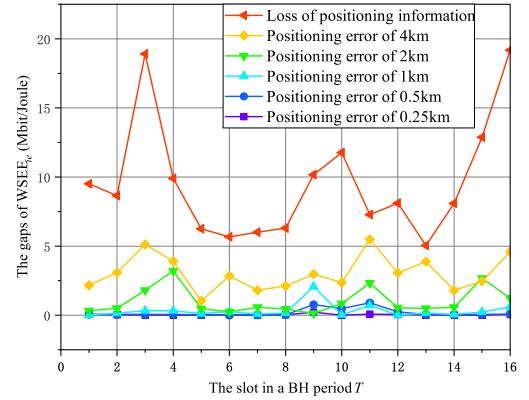


Fig. 16. Gaps of WSEE between perfect positioning and positioning errors in a BH period T , where scale factors of traffic demand are 1 and maximum transmit power is 3 dBW.

Fig. 15, the gaps are always greater than 0. The effectiveness of the proposed algorithm in maximizing the WSEE is verified. Meanwhile, within a 1 dB SINR loss, the impact on the algorithm is relatively minor.

When the TTs access the satellite, they report their positioning information [43]. However, due to the positioning error and the mobility of some TTs (e.g., IoV and ships), it leads that the channel gain contains errors. Thus, to evaluate the performance of the proposed method when positioning information contains errors, we compare the WSEE of BH patterns in different positioning errors, even when there is no positioning information, we assume the terminals are located at the beam center. The results are shown in Fig. 16.

It shows the gaps between perfect positioning and positioning errors in a BH period, and the horizontal coordinates indicate the beam pattern at each slot. The gap can be represented by

$$\Delta WSEE_{ie} = WSEE_p(F_p, w_p) - WSEE_p(F_e, w_e) \quad (45)$$

the subscripts p and e of these two variables indicate that the results are calculated by proposed JPFA algorithm under the perfect positioning conditions and positioning error conditions. $WSEE_p()$ denotes the scenario with perfect location

information. It can be observed that as the location error increases, the error of WSEE in each time slot also gradually grows, and all errors are greater than zero. This further demonstrates the robustness of the proposed JPFA algorithm in maximizing WSEE. Within an error range of 2 km, only a relatively minor impact is incurred. Meanwhile, the impact of the positioning error on WSEE further demonstrates the suitability of JPFA for dynamic mobile environments. For instance, IoV terminals traveling at typical urban speeds (e.g., ≤ 100 km/h) and under the assumption that the TTs report their positions once every 100 BH periods [43], undergo a maximum cumulative positioning error of 88 m, which remains within the tolerance range (< 2 km positioning error). Additionally, the suitability of JPFA ensures sustained WSEE optimization without recomputation per BH period.

V. CONCLUSION

In this article, we have investigated the joint uplink resource allocation for LEO satellite BH systems, and we also have considered the fact that different types of terminals are not uniformly sensitive to EE. Thus, a resource optimization problem has been formulated to maximize the WSEE by uplink BH patterns design, jointly optimizing power control and frequency allocation. Then, to solve this NP-Hard problem, it has been decomposed into two subproblems. First, we have built a “friendship” network based on distance and EES to describe the peer effect among beams. To balance the complexity, we have proposed two BBHMA methods according to many-to-one matching to obtain BH patterns. Second, by considering the EES of different TT types, we have proposed a JPFA method to jointly allocate resources in the frequency and power domains, maximizing the EE of system while reducing inter-TT interference and satisfying traffic demands. Finally, simulations have proved that the proposed BBHMA and JPFA methods have performed better in EE than benchmarks. Moreover, the applicability of these algorithms in dynamic terminal scenarios has been further validated. This work provides a novel paradigm for uplink resource management.

For future work, we will further explore leveraging multi-satellite cooperation for uplink resource management and developing a dynamic trajectory model on the terminal side to evaluate the mobility impact on resource management.

APPENDIX

The optimization result of the primal $\mathcal{P}_3$ is p^* , $p^* = \max_{\mathbf{w}, \mathbf{F}} E_{\text{total}}(\mathbf{w})$, and the optimization result of lagrangian dual problem is d^*

$$d^* = \min_{\zeta} \max_{\mathbf{w}, \mathbf{F}} L(\mathbf{w}, \mathbf{F}, \zeta) \quad (46)$$

according to weak duality theorem, we can get the following inequality:

$$p^* = \max_{\mathbf{w}, \mathbf{F}} E_{\text{total}}(\mathbf{w}) \leq d^* = \min_{\zeta} \max_{\mathbf{w}, \mathbf{F}} L(\mathbf{w}, \mathbf{F}, \zeta). \quad (47)$$

We can obtain 2 cases of $\mathbf{w}$ and $\mathbf{F}$ in accordance with the constraints $C4'$, $C1''$, $C2_a$, $C2_b$, $C6_a$.

Case 1: When the optimal solution satisfies $\sum_{\forall j} \sum_{l \in L_j^i} \sum_{m=1}^M (f_m^{j,l} - (f_m^{j,l})^2) = 0$, the $f_m^{j,l}$ is obviously an integer solution which satisfies the $\mathcal{P}_3$. Then we assume ζ^* is the optimal value, we can get

$$p^* = \max_{\mathbf{w}, \mathbf{F}} E_{\text{total}}(\mathbf{w}) = d^* = \min_{\zeta} \max_{\mathbf{w}, \mathbf{F}} L(\mathbf{w}, \mathbf{F}, \zeta^*). \quad (48)$$

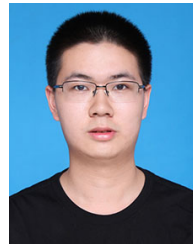
Furthermore, $L(\mathbf{w}, \mathbf{F}, \zeta)$ is a monotonically decreasing function with respect to ζ , it means that for any $\zeta \geq \zeta^*$, the $p^* = d^*$ in case 1, i.e., $\mathcal{P}_4$ is equivalent to the $\mathcal{P}_3$.

Case 2: When the optimal solution is not binary, i.e., $\sum_{\forall j} \sum_{l \in L_j^i} \sum_{m=1}^M (f_m^{j,l} - (f_m^{j,l})^2) > 0$, the d^* tends to $-\infty$ in this case as long as ζ is large enough. However, this may be impossible to occur since the solution of $\mathcal{P}_3$ is always greater than 0. Thus, the optimal solution must be satisfied $\sum_{\forall j} \sum_{l \in L_j^i} \sum_{m=1}^M (f_m^{j,l} - (f_m^{j,l})^2) = 0$, and while ζ is large enough, the $\mathcal{P}_4$ is equivalent to the $\mathcal{P}_3$.

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